Nippon AI DOJO
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FOIL STAMP ANOMALY DETECTION

Final Report

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Agenda

01	Project Overview
02	Progress
03	Results
04	Challenges
05	Limitations
06	Summary & Suggestions

Client Business & Problem

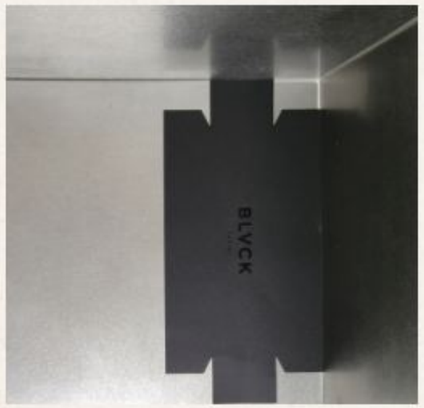
01 Client's Business

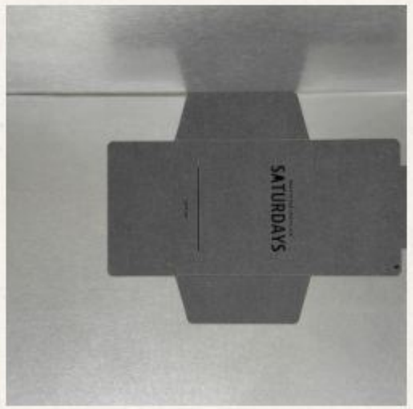
Foil stamping service using a metal plate and heat-pressed foil film to produce high-quality premium finishes on paper and other materials.

02 Client's Motivation

High printing speed makes early misprints hard to notice, and the defect continues until the job finishes, causing losses.

- **Wasted labor**
- **Wasted materials**
- **Waste of time**









Overview

Foil stamping service using a metal plate and heat-pressed foil film to produce high-quality, premium finishes on paper and other materials.

Proposed Solution

Develop " *few-shot AI model* " to find anomalies in paper printed by foil stamping.

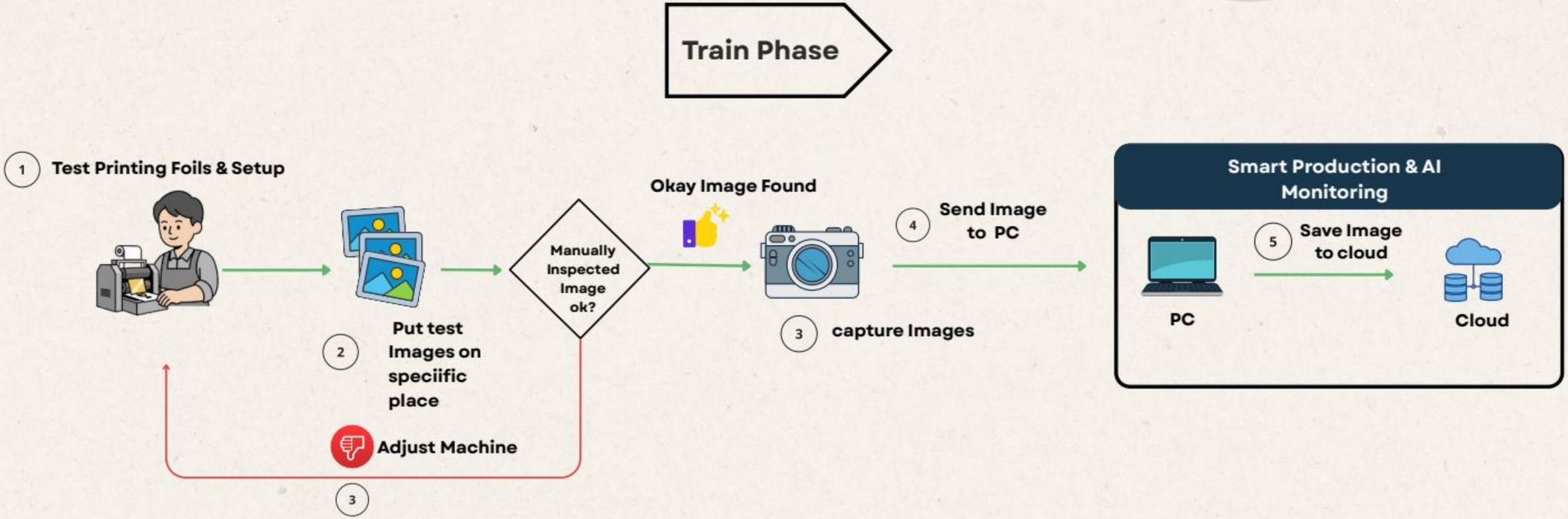


Figure: Workflow of foil stamping Model Training Phase

Proposed Solution

Develop AI model to find anomalies in paper printed by foil stamping.

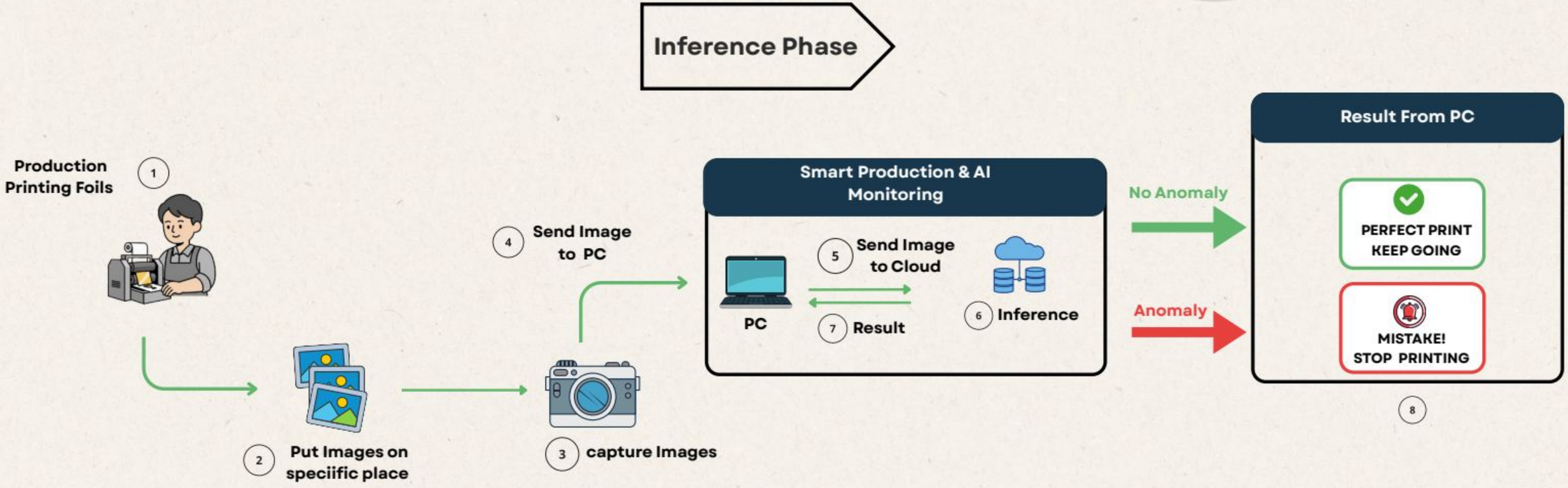


Figure: Workflow of foil stamping anomaly detection

Schedule and Data

Phase-1
- Find dataset
- Class selection of the dataset

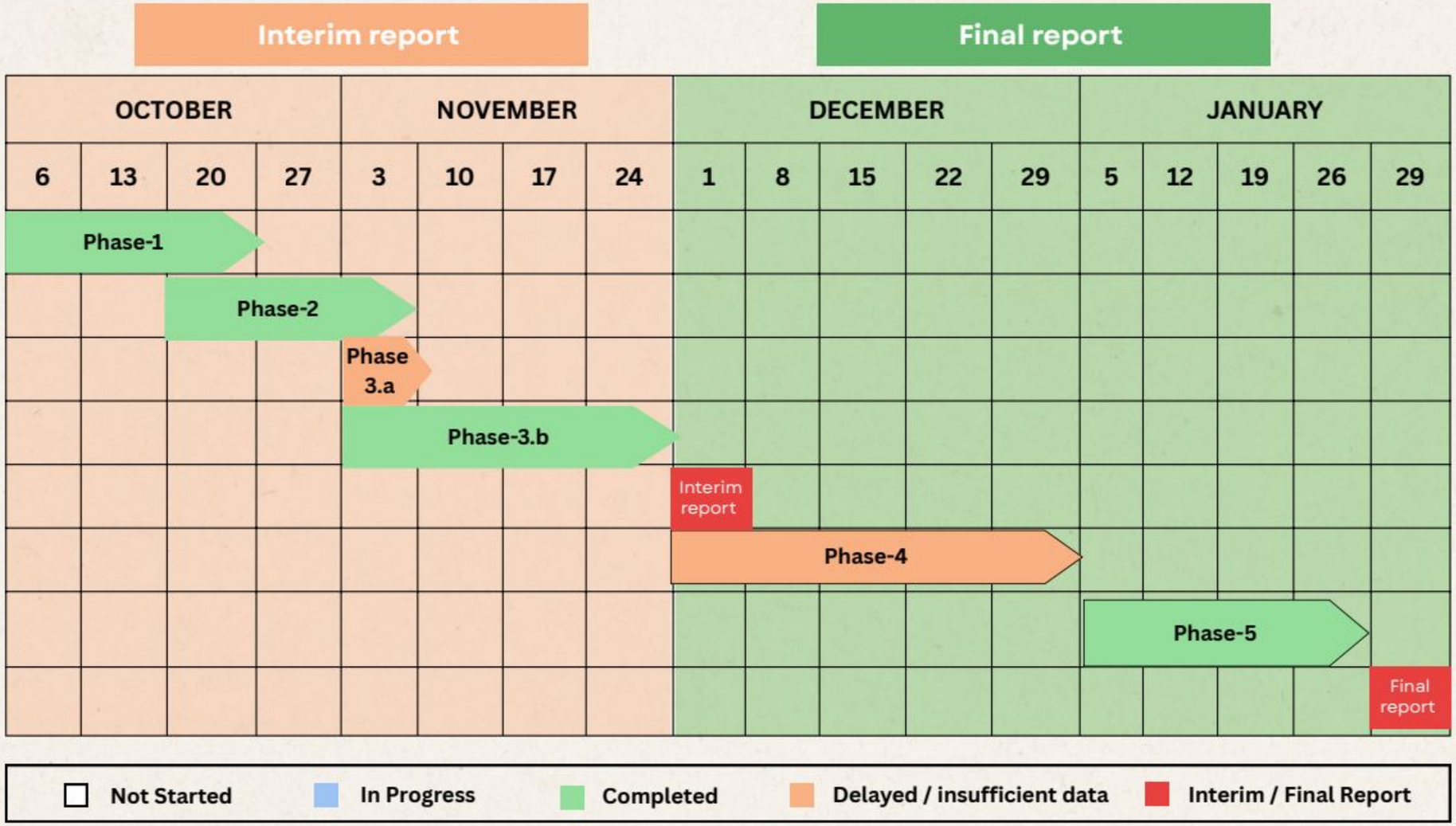
Phase-2
- Model Setup
- Development on temporal data

Phase-3.a
- Development Data

Phase-3.b
- Development on development data

Phase-4
- Got new data.(insufficient)
- Improve the model

Phase-5
- Select Best model
- Make final Report



Timeline of receiving data

Sample Data
October-6

Additional Data
November-3

Sample-4k
November-10

Development Data(1)
November-19 and 25

Development Data(2)
December 4 and 29



Figure: Sample Data



Figure: Development Data(1)



Figure: Development Data(2)

Data Description

- The dataset contains very limited real anomalous samples
- Limited anomalies make reliable and fair model evaluation challenging
- To address this, we introduced small synthetic anomalies during evaluation
- Synthetic anomalies enable robust, consistent, and comparable evaluation



Figure: White circular regions filled

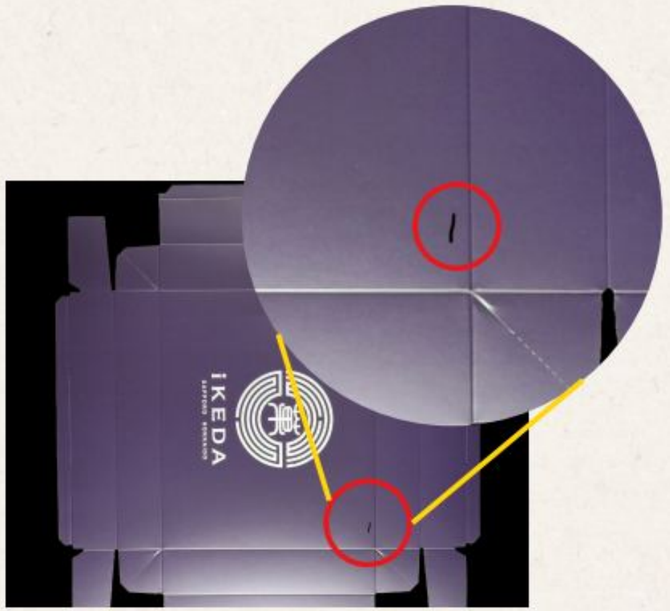


Figure: Black color dot

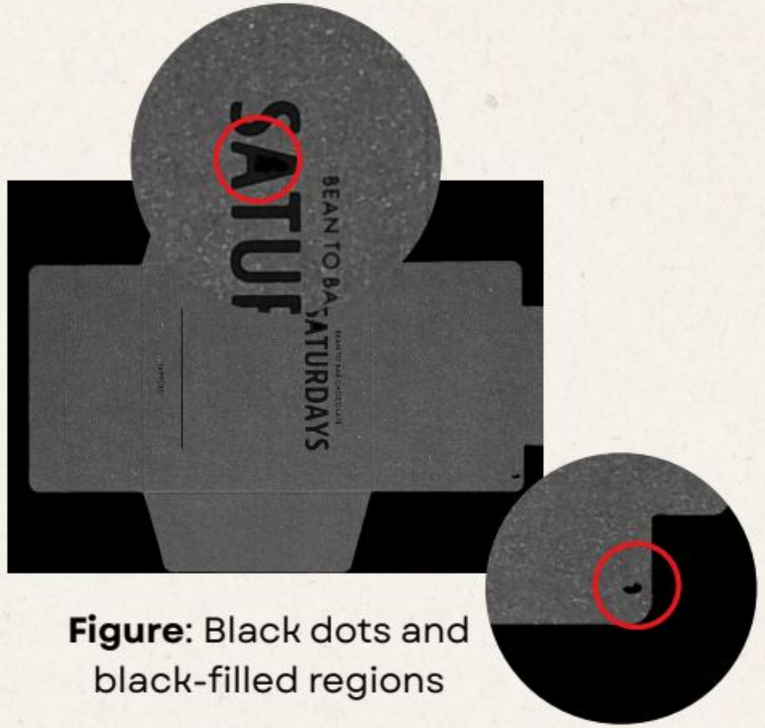
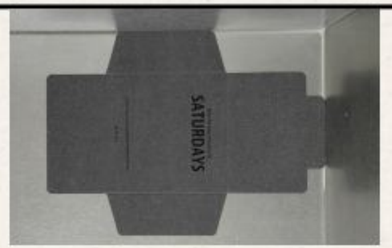


Figure: Black dots and black-filled regions

Data Description

Class Image	Class Name	No. OK image	No. Not Ok
	SATURDAYS (with 1 synthetic anomaly)	1	1
	Rikichi Candle	1	1
	sgd	4	2

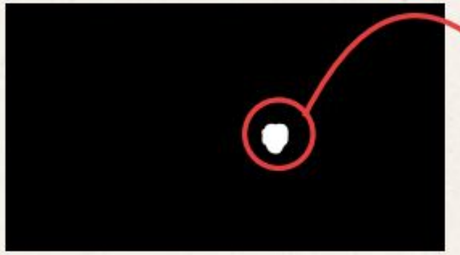
Class Image	Class Name	No. OK image	No. Not Ok
	Ikeda shokuhin (with 2 synthetic anomaly)	4	3
	NY	4	1

Experimentation with Data

4 **Training the Model**

Preprocessed data sent to the model for training

3 **Annotate Data & Segmentation Mask**



Indicates anomaly

2 **Image Alignment & Perspective Transformation**



1 **Remove Background**



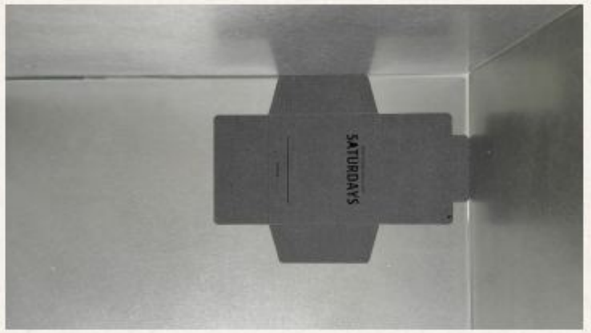
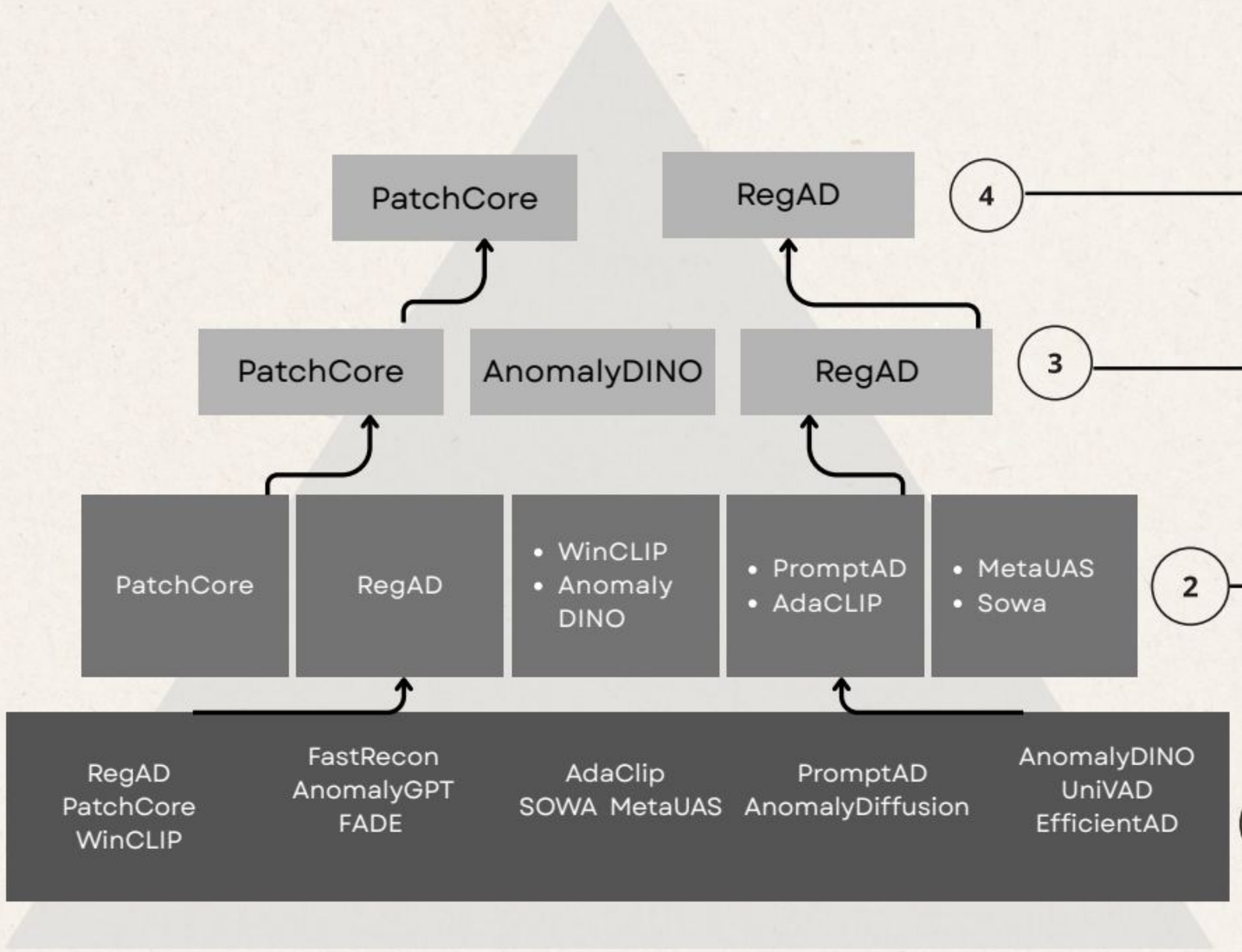


Figure: Original Data

Experimented Models



Final Models

Among the 5 shortlisted models, **3 models were performing the best - Till Interim**

From these 14 models, we **narrowed the list down to 5 models** based on few-shot adaptation capability, accuracy, and speed.

Initially selected **14 models** as potential candidates

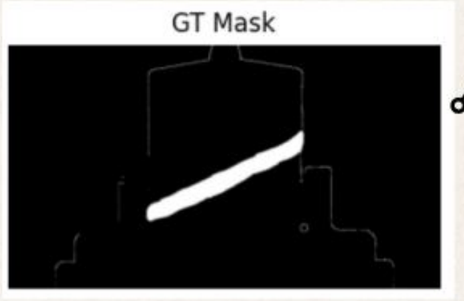
RegAD



Support Image/ Train Image



Test Image



Ground Truth Mask of anomaly image

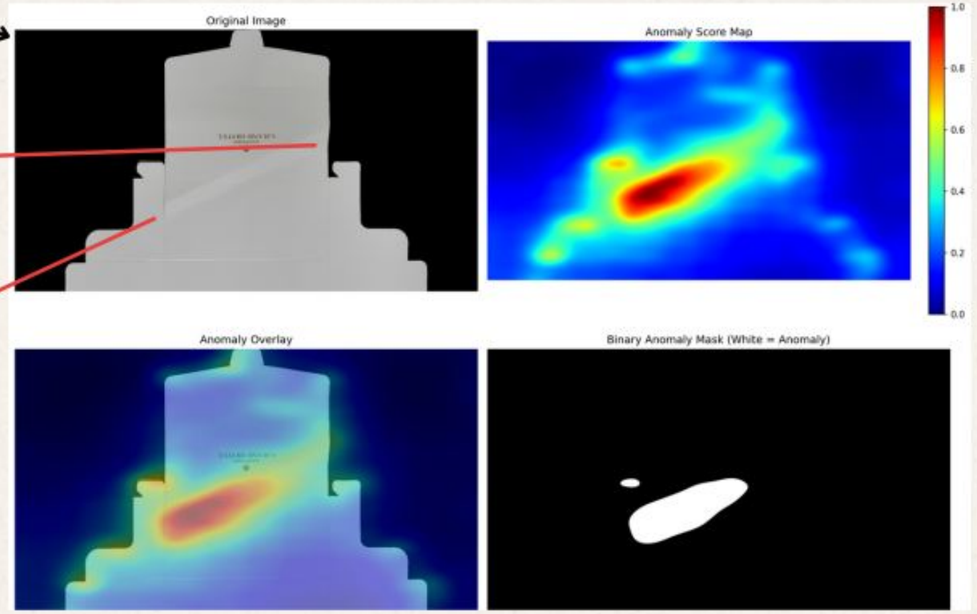
RegAD Model

Anomaly Threshold: 0.05

Image Name	Anomaly Score	Prediction
sgd_NG_1.jpg	0.0890	❌ ANOMALY
sgd_NG_2.jpg	1.0000	❌ ANOMALY
sgd_OK_4.jpg	0.0000	✓ NORMAL

Classification output





Visualization output

Results - RegAD

RegAD results for all classes

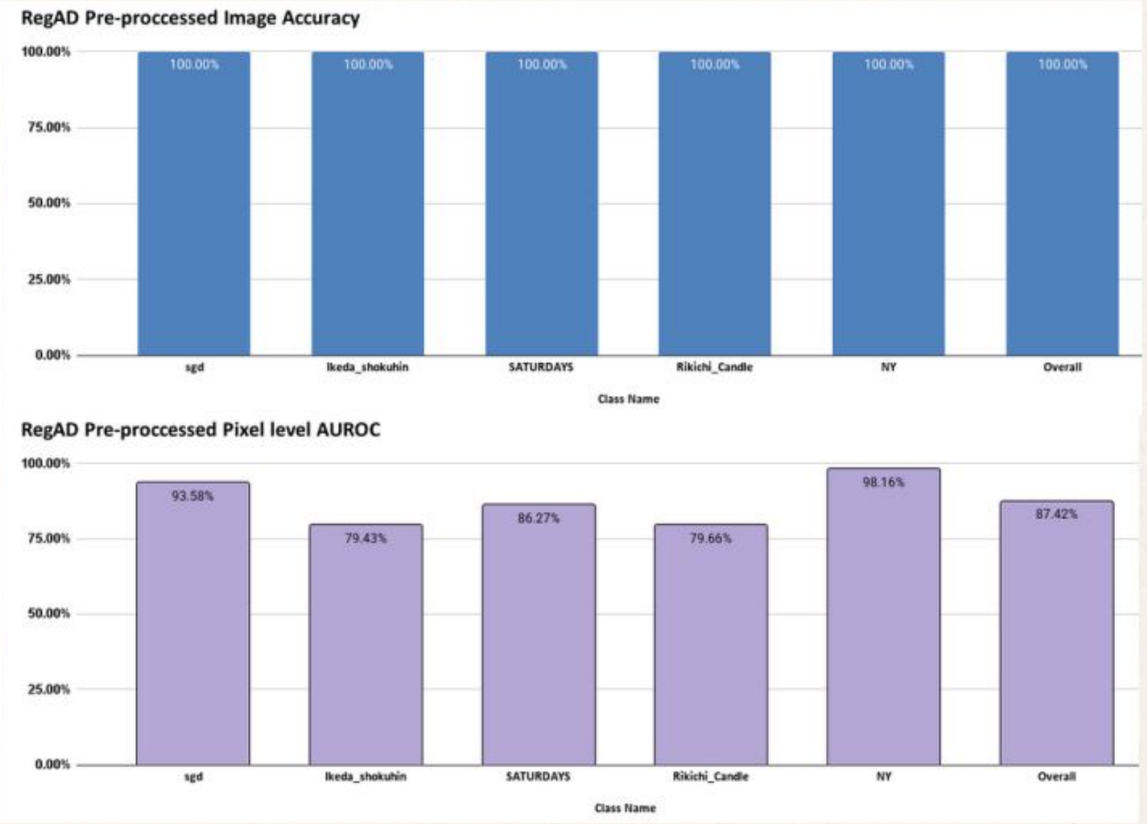


Figure: RegAD Accuracy and Pixel-level AUROC

Metrics of Pre-proccesed data

Classification Accuracy(Synthetic and Original data): **100%**
Mean Pixel level AUROC: **87.42%**
Inference Time: **~1.1 seconds (each class)**

	Raw client data	Pre-processed data
Classification Accuracy	93%	**100%**
Mean_Pixel_AUROC	69.15%	**87.42%**

Figure: Comparison between Raw client data and pre-processed data through our pipeline

PatchCore

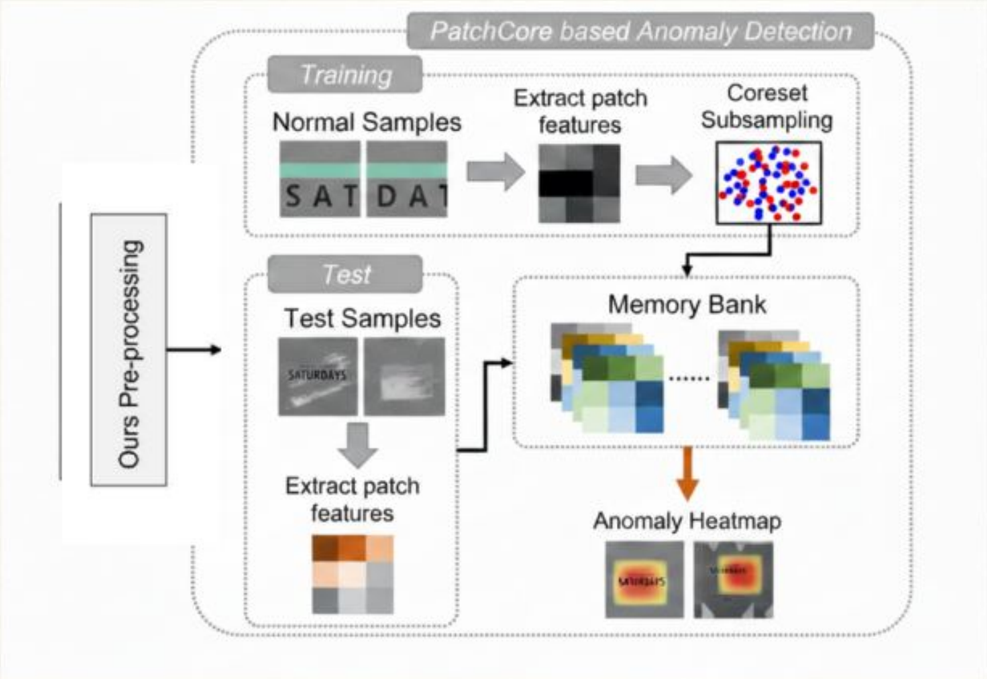
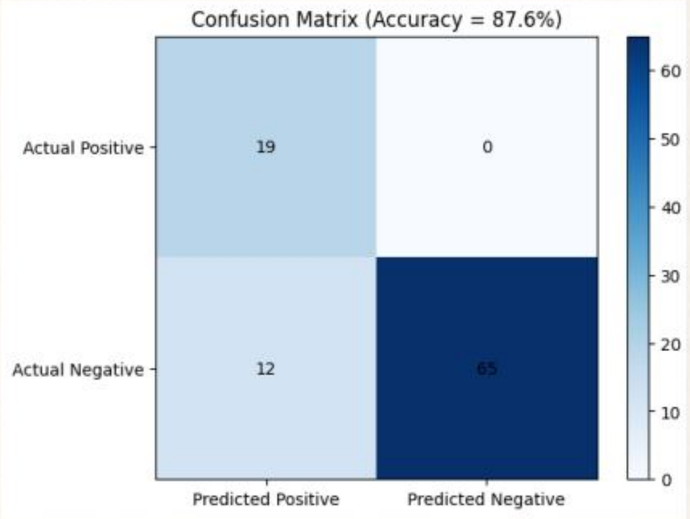


Figure: PatchCore-based Anomaly Detection Architecture



- Classification Accuracy: **87.6%**
- Mean_Pixel_AUROC: **63.57%**
- Inference Time: ~**1.2** seconds per image

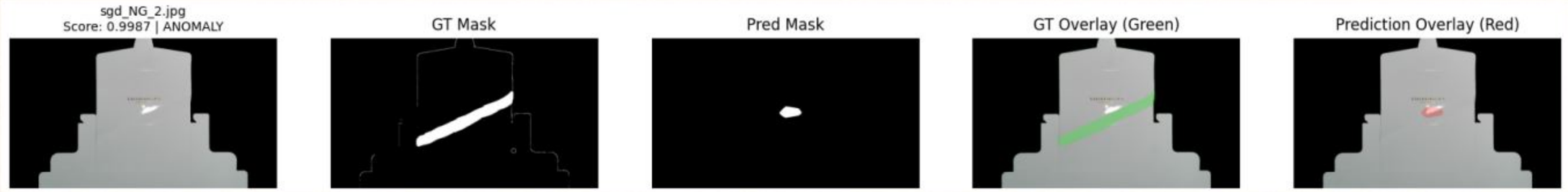
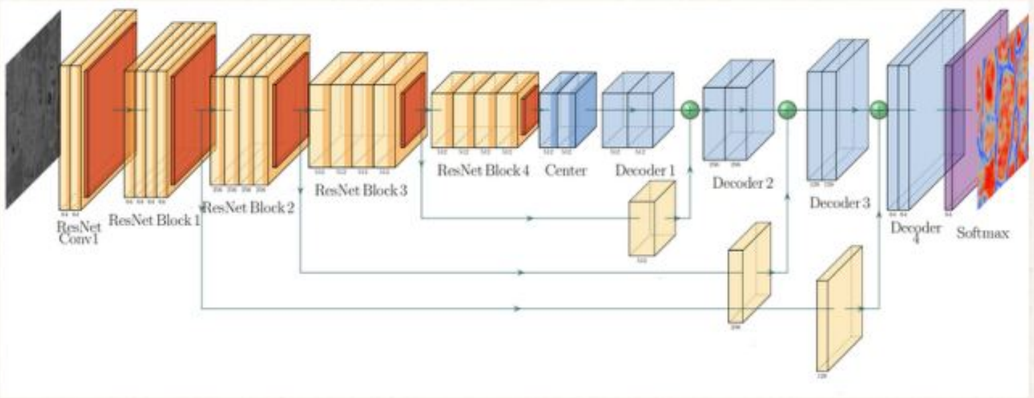


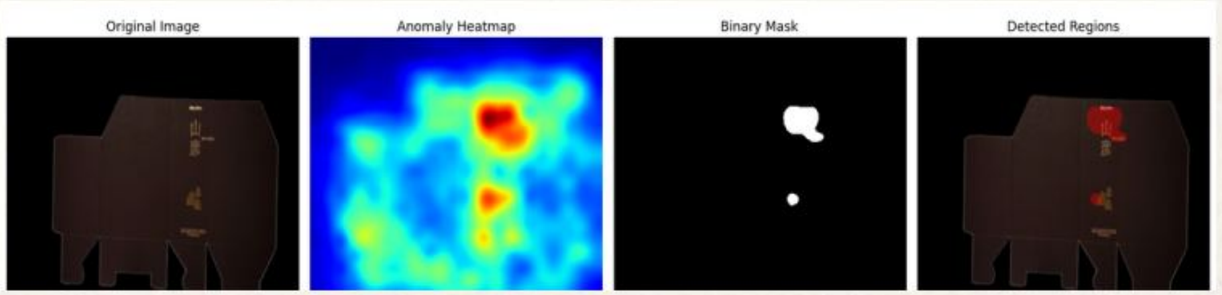
Figure: Test Image Mask and Overlay

Enhanced : PatchCore

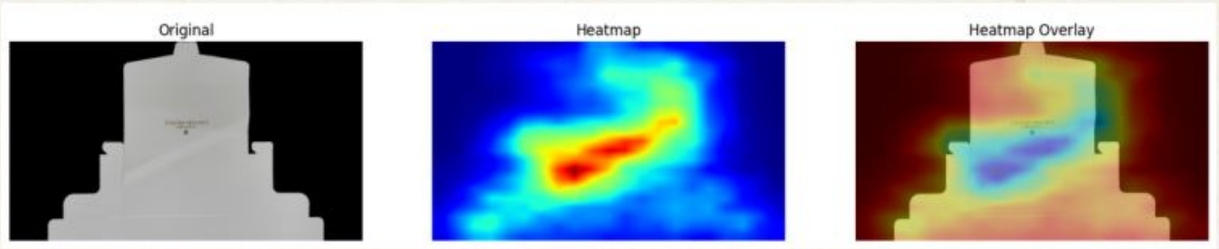
- BACKBONE = "**resnet18**"
- FEATURE_LAYER_INDEX = **3**



- Set Patch Number to **2000**
- Integrated **Adaptive Thresholding**



- Integrated **Overlay and Heatmap** On Anomaly Region



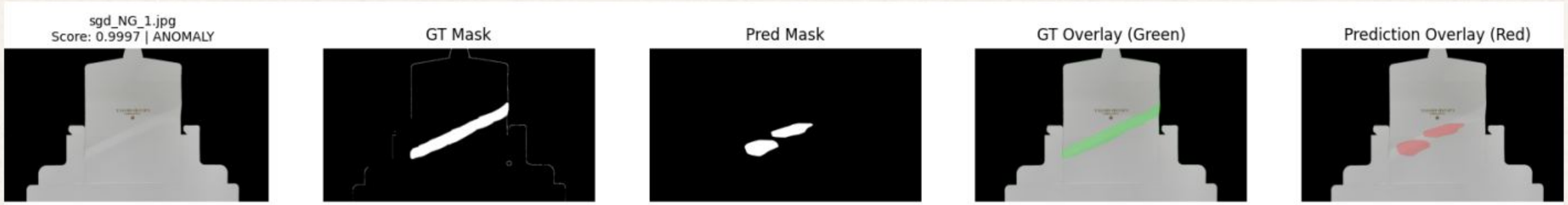


Figure: Test Image Mask and Enhanced model Overlay

Results - PatchCore

	Base Architecture	**Enhanced Architecture**
Classification Accuracy	87.6%	**100%**
Mean_Pixel_AUROC	63.57%	**83.02%**
Adaptive Thresholding	Not Present	**Present**
Overlay over Anomaly Region	Not Present	**Present**
Inference Time (seconds)	~1.2	**~0.013**

Patch-Core results for all classes

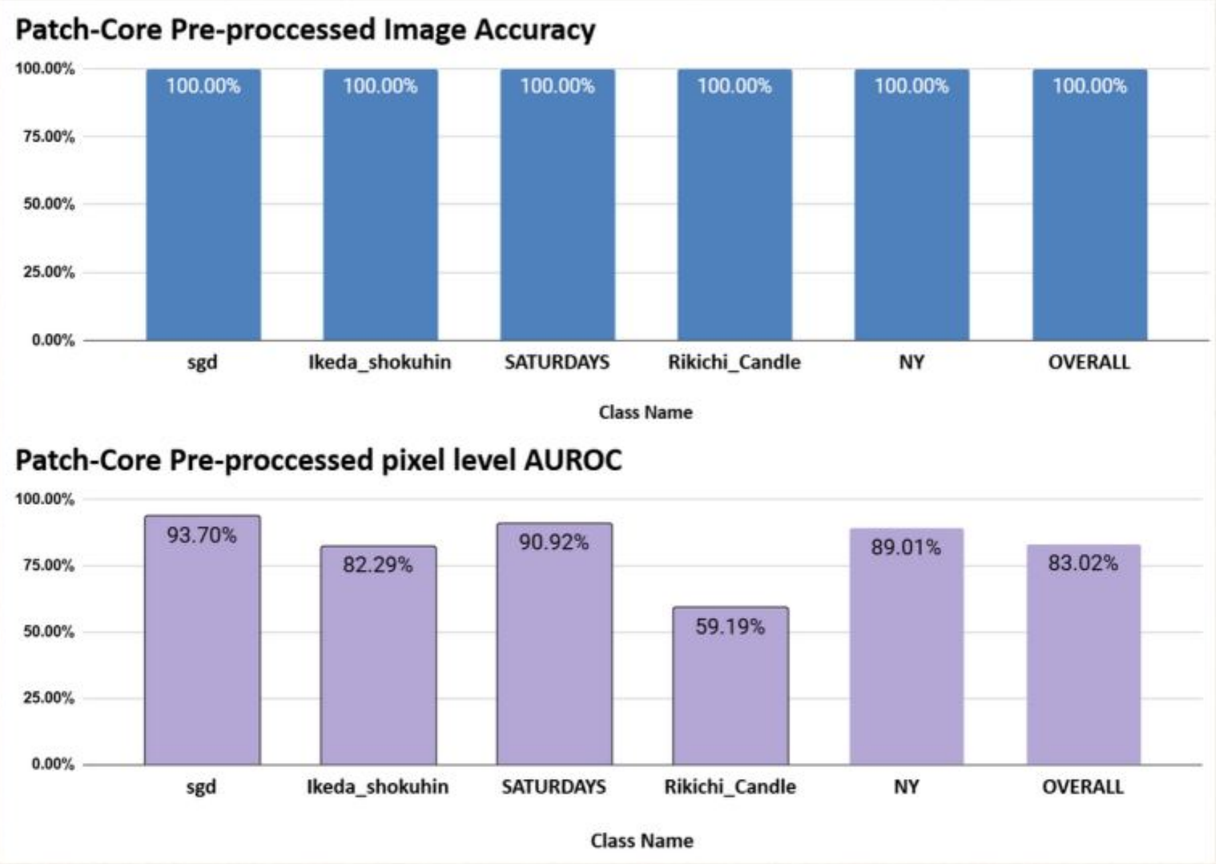


Figure: **PatchCore** Accuracy and Pixel-level AUROC

Model Comparison

	RegAd	PatchCore
Classification Accuracy	**100%**	**100%**
Mean_Pixel_AUROC	**87.42%**	83.02%
Adaptive Thresholding	Not Present	**Present**
Overlay over Anomaly Region	**Present**	**Present**
Inference Time (seconds)	~1.1	**~0.013**

Challenges & Solutions

3.a Issues We Faced - insufficient ‘Okay’ Images

- The number of Support image was low to train the model

3.a Solution - image augmentation

- Applied image augmentation (e.g. changed **brightness hue, contrast** to make multiple ‘ok’ images)



Figure: Original Image

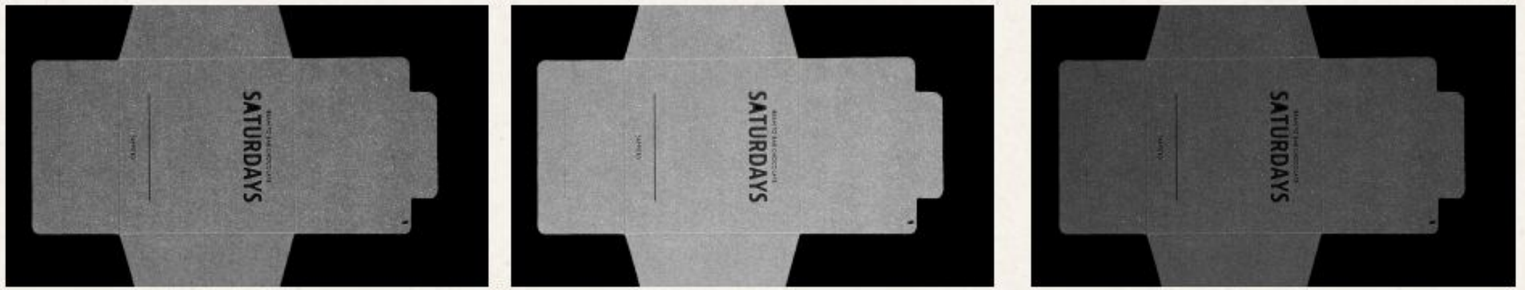


Figure: Changed Contrast, Brightness, Hue

Challenges & Solutions

3.b Issues We Faced - Environmental

- Lots of Unnecessary Information

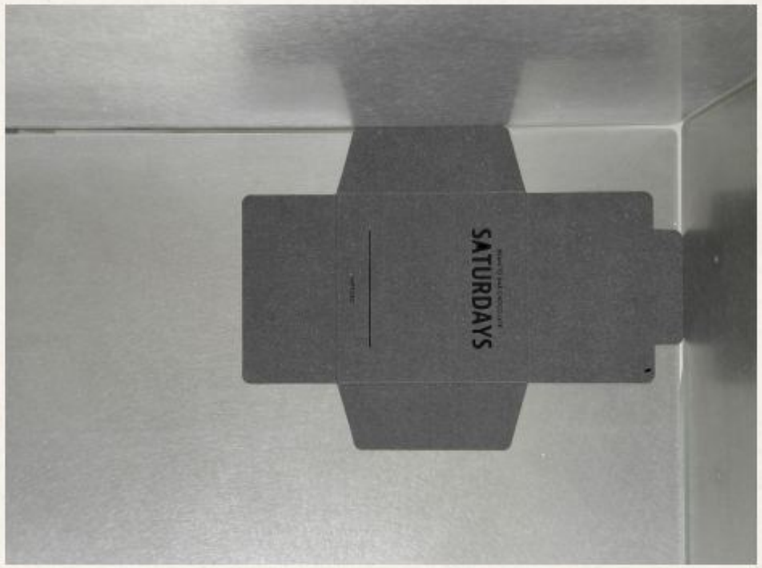


Figure: Original Image

3.b Solution - Advanced BirefNet

- We used a pretrained Background Removal Model - Birefnet

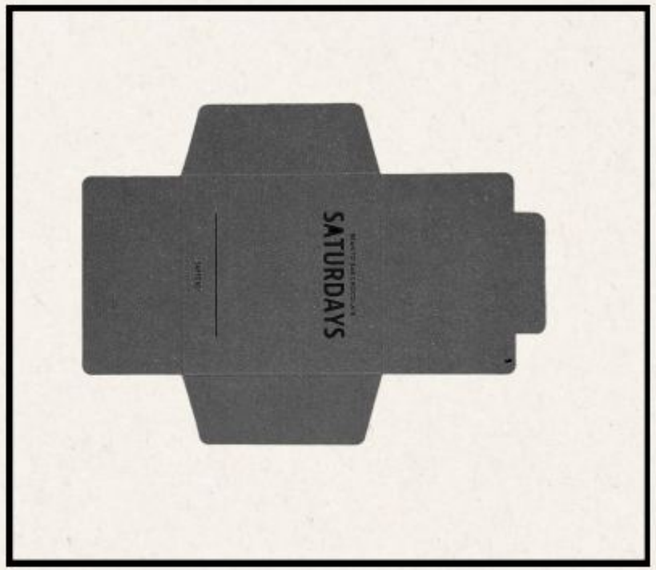


Figure: Background Removed Image

Challenges & Solutions

3.c Issues We Faced - Size, Shape, Alignment

- Image gets scaled after background removal.
- Images have varying orientations, causing inconsistencies.

3.c Solution - Transformation

- We resize the image, put a background and placed it on the center
- We align the image using **perspective transformation.**

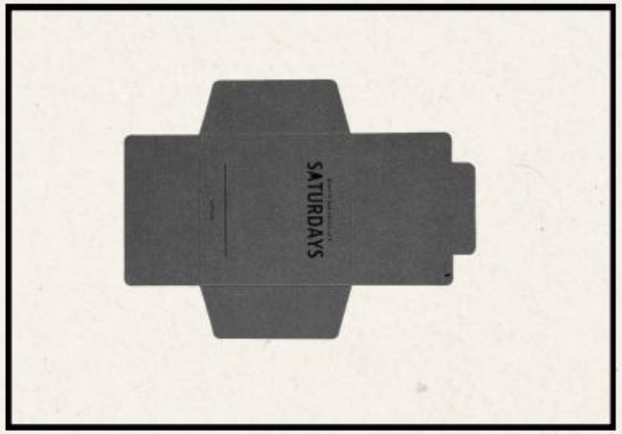


Figure: Scaled Image





Figure: Final Transformed Image

Challenges & Solutions

3.d Issues We Faced- Drift / Shift

- Even after preprocessing, the model failed, due to new type of anomaly, after getting development data. (Drift / Shift of logos, fonts)

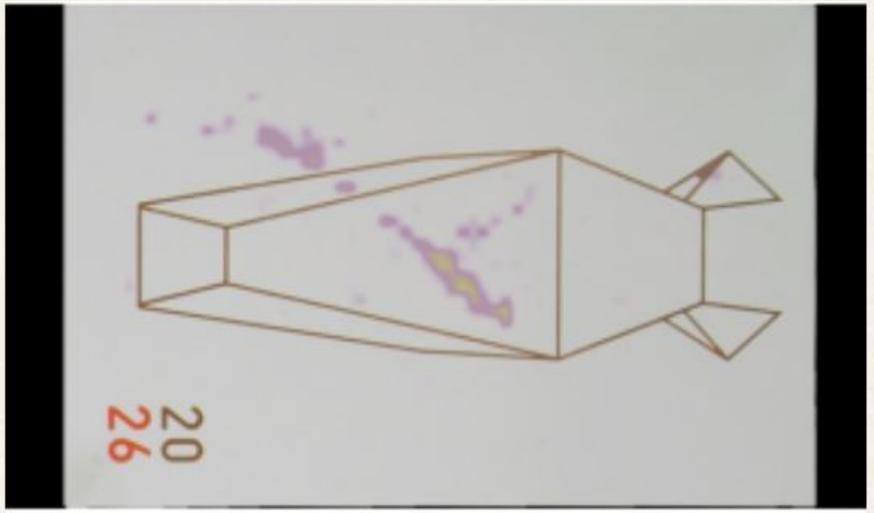


Figure: Failed to detect Anomaly

3.d Solution- ORB

- ORB feature-based matching combined with homography estimation

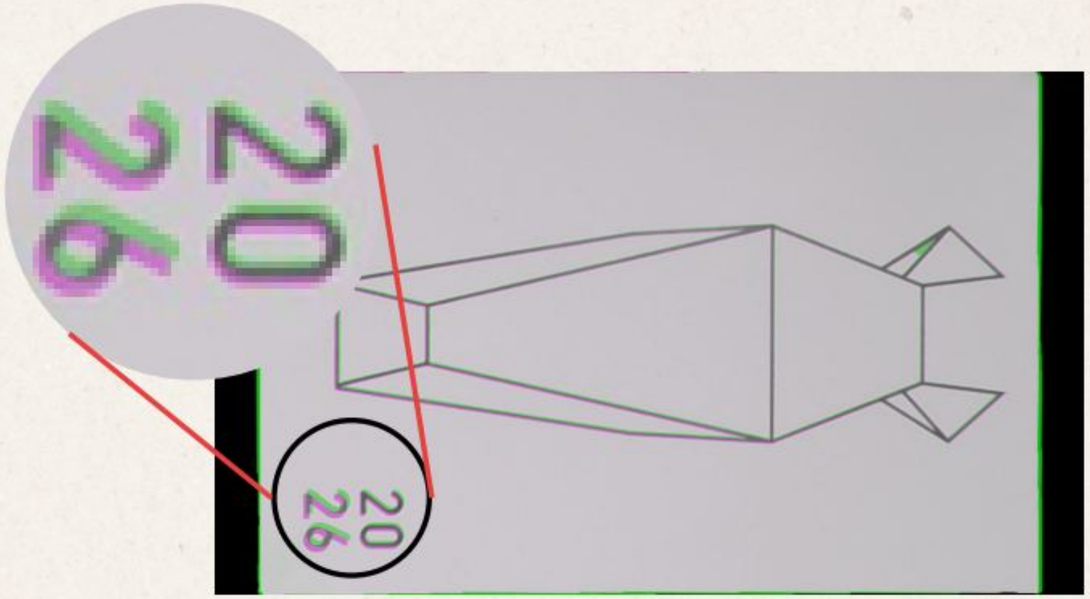


Figure: Drift Detection

Value Delivered

- Detect scratch
- Detect wrinkles
- Misplaced Stamps
- Detect foil logo, font drift
- Detect unintended spots, lines, etc.

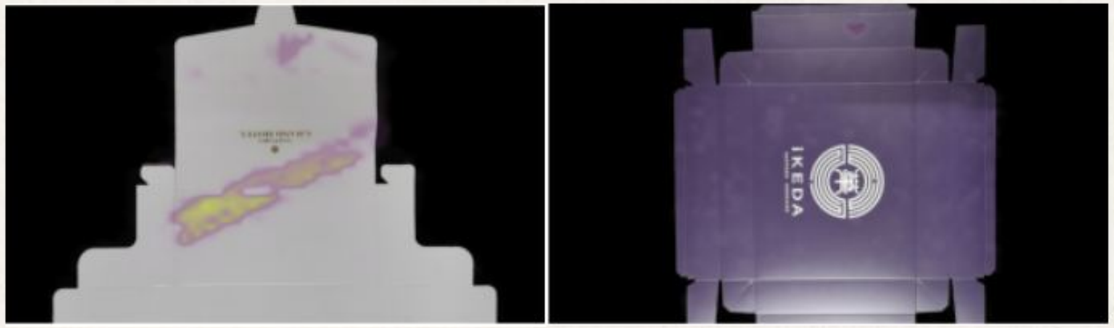


Figure: Detected Anomaly

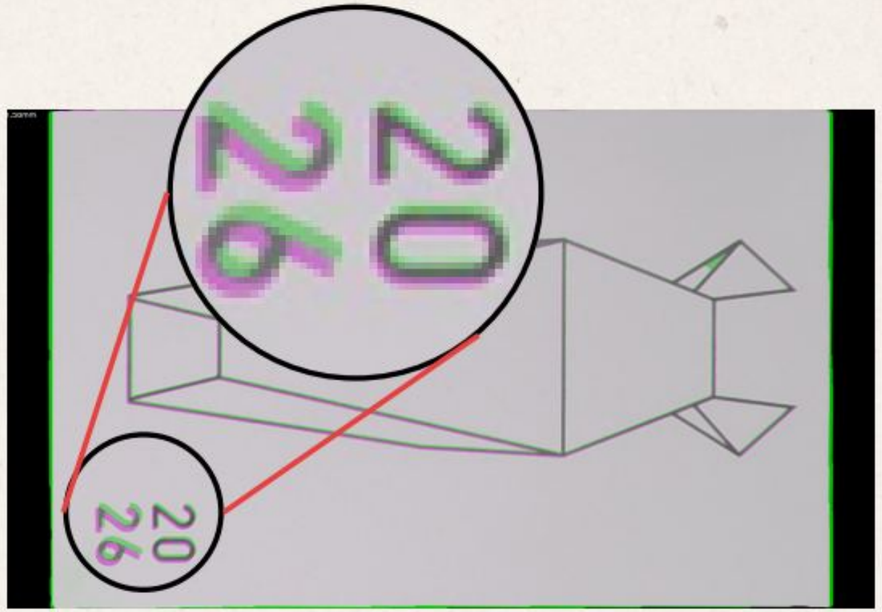


Figure: drift

Limitations - Models

- Failed to detect small spots or minor anomalies
- Failed to remove some part of the Background precisely
- Incorrect detection of anomalies

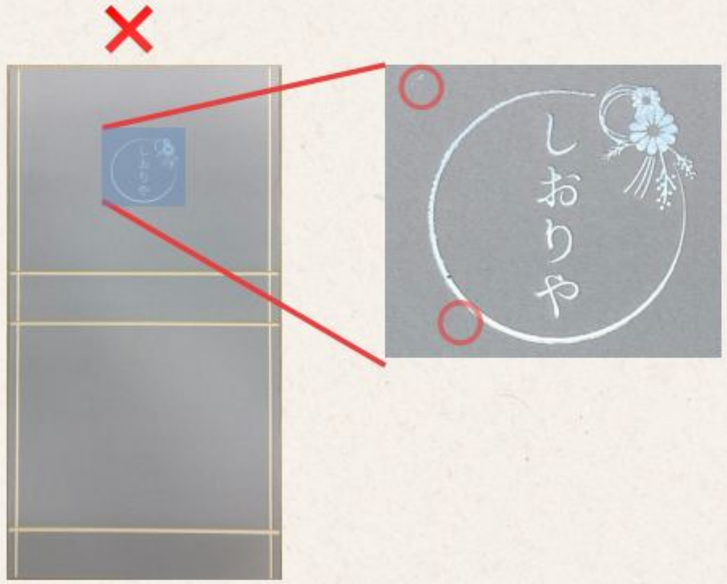


Figure: Fail to detect small anomaly

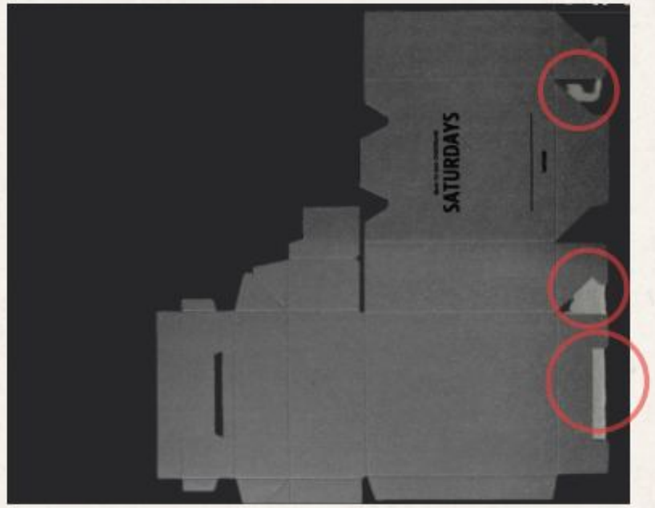


Figure: Fail to remove background properly

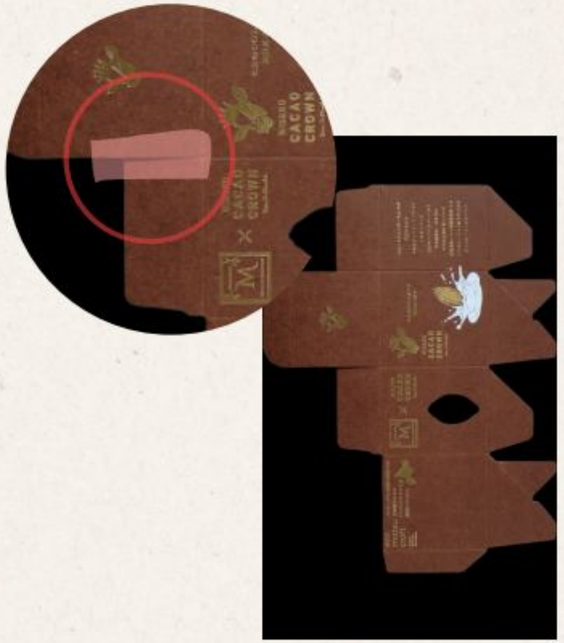


Figure: Wrong prediction due to okay image

Limitations - Client data

- **Inadequate data** and proper okay images
- Poor reflection and shadow on the foil-stamped surface



Figure: Poor reflection on images



Figure: Unnecessary shadow due to uneven surface of foil

Summary & Suggestion

- Supplement the dataset with more OK/NG images for better experimentation
- Take measures to minimize surface reflections
- Optimize the environment to reduce distracting shadows from the foils
- Ensure proper camera allignment while capturing images
- Finetune the BirefNet architecture for improved performance

Thank You